A Simulation Study of CNN-Based Cooperative Spectrum Sensing Using Binary Local Decisions

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Abstract

Cooperative spectrum sensing combines observations from multiple secondary users to detect primary-user activity in a cognitive radio network. This paper presents a course-project reimplementation of the convolutional neural network (CNN) sensing component of Do and Koo's framework, restricted to binary decision fusion. A two-state Markov process generates primary-user activity, and normalized Gaussian energy statistics generate local decisions under variable signal-to-noise ratio (SNR). A compact CNN is trained with bipolar input encoding, Adam optimization, and validation-based checkpoint and restart selection. The archived MATLAB experiments compare the CNN with strict-majority fusion and single-user energy detection, and examine the effects of the number of cooperating users and the nominal sensing sample count. At a mean SNR of -12 dB, with ten users and 300 sensing samples, the CNN achieves a sensing-error sum of 0.3254, compared with 0.3857 for strict majority and 0.7025 for one user. However, a diagnostic audit shows that the CNN differs from at-least-half voting on only two of 1,024 binary input patterns. The results support the benefit of cooperation within the simulated model, while showing that the apparent advantage over strict majority is closely associated with tie handling and does not establish a general advantage over voting.

Index Terms

Cognitive radio, cooperative spectrum sensing, convolutional neural network, energy detection, binary decision fusion, simulation study.

I. Introduction

Spectrum sensing enables a secondary user (SU) to assess whether a licensed channel is occupied by a primary user (PU). A missed detection can expose an active PU to secondary interference, whereas a false alarm rejects an available transmission opportunity. The sensing problem therefore requires simultaneous consideration of detection and false-alarm probabilities. Cooperative spectrum sensing collects local sensing outcomes at a fusion center (FC), allowing several observations of the same PU state to inform one global decision [1]–[4].

Do and Koo [1] developed a broader cognitive-radio framework that combines CNN-based cooperative spectrum sensing with energy harvesting and a transfer-learning actor-critic policy for protected data transmission. Their sensing component uses binary decisions from multiple SUs as the CNN input. The present project studies this component as a self-contained supervised classification problem. Energy harvesting, battery evolution, encryption, transmission scheduling, and reinforcement learning are outside the implemented scope. Consequently, sensing results in this paper are not interpreted as measurements of throughput, security, or energy efficiency.

The project retains the binary local-decision representation, the normalized Gaussian approximation for energy detection, and the compact convolution-ReLU-fully-connected architecture of [1]. It modifies the input encoding, optimization method, data-generation budget, and validation protocol. Within the course timeframe, the changes deliberately prioritize a carefully documented and assessable implementation of the sensing component. They are not presented as a new sensing algorithm or an exact reproduction of all original experimental settings.

The study addresses three questions: how the implemented CNN compares with a fixed voting rule and one local detector; how its sensing-error metric changes with the number of cooperating SUs; and how that metric changes with the nominal number of samples in the local energy model. Interpretation of the first question requires particular care. The code labels its voting baseline `HALF`, but implements strict majority. In contrast, the half-voting rule described in [1] accepts a tie as an active-PU decision. The supplied diagnostic results are therefore used to distinguish the benefit of cooperation from the effect of the tie policy.

The practical outputs of the project are an explicit simulation and learning pipeline, three controlled parameter studies, and an analysis of what the learned binary decision rule actually does. Section II defines the system and local detector. Section III describes the CNN and training protocol. Section IV presents the simulation setup, archived results, and their limitations. Section V concludes the study.

II. System Model and Local Sensing

A. Primary-User Activity and Cooperative Observation

Consider a cognitive radio network (CRN) consisting of a licensed primary transmitter (PU-Tx), a primary receiver (PU-Rx), K secondary users (SUs), and a fusion center (FC), as conceptually illustrated in Fig. 1. The network topology is adapted from the cooperative spectrum-sensing framework in [1], with the present study restricted to primary-user activity detection

and binary decision fusion. The primary link between PU-Tx and PU-Rx is included to illustrate the licensed communication environment; however, its transmission and reception processes are not explicitly simulated.

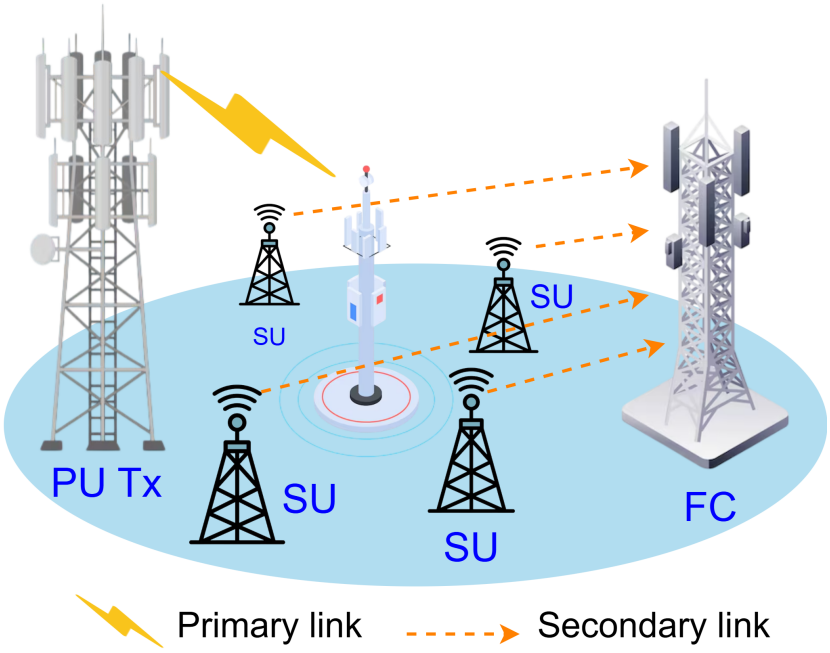


Fig. 1. Conceptual system model of the considered CNN-based cooperative spectrum-sensing network, adapted from [1]. The primary link provides the licensed-channel context, PU-to-SU sensing links represent statistical sensing relationships, and reporting links convey one-bit local decisions to the FC.

The system operates in discrete time slots. At each slot, the PU is either active or inactive, and its activity state remains unchanged throughout the sensing interval. Each SU independently performs local energy detection and produces a one-bit sensing decision $D_{i,t} \in \{0, 1\}$, where $i \in \{1, \ldots, K\}$ denotes the SU index and t denotes the time-slot index. These local decisions are assumed to be conveyed to the FC through ideal reporting links. The FC combines the reported decisions using a trained CNN to estimate the PU activity state.

In the implemented simulation, the PU-to-SU sensing links are represented through SNR-dependent Gaussian energy statistics rather than explicitly generated waveforms or fading-channel coefficients. Reporting errors and delays are neglected, while the primary receiver and subsequent secondary data transmissions are not modeled. Accordingly, the resulting global decision is used solely to evaluate the sensing performance of the considered fusion methods.

Let $H_t \in \{0, 1\}$ denote the true PU activity state in slot t, where $H_t = 0$ and $H_t = 1$ correspond to inactive and active PU states, respectively. Following the two-state Markov activity model in [1], the implemented transition matrix is

$$\mathbf{P} = \begin{bmatrix} 1 - p_{01} & p_{01} \\ p_{10} & 1 - p_{10} \end{bmatrix} = \begin{bmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{bmatrix}. \tag{1}$$

Rows and columns correspond to the previous and next states, respectively. The stationary probabilities follow directly from the two-state Markov chain balance equations, which satisfy $\boldsymbol{\pi}\mathbf{P} = \boldsymbol{\pi}$ and $\pi_0 + \pi_1 = 1$, giving

$$\pi_0 = \frac{p_{10}}{p_{01} + p_{10}} = \frac{1}{2}, \qquad \pi_1 = \frac{p_{01}}{p_{01} + p_{10}} = \frac{1}{2}. \tag{2}$$

The generator draws the first state from this stationary distribution. Finite sequences need not contain exactly equal class counts. Each SU reports one bit per slot to the FC; reporting errors, losses, and delays are not modeled. The CNN uses only the current slot's bits, so it does not exploit the temporal history of the Markov process.

B. Signal Interpretation and Simulated SNR

The local detection problem uses the classical energy-detection principle [5], [6] and is motivated by the received-signal and energy-statistic definitions in [1, eqs. (5)–(6)]:

$$x_i[n] = \begin{cases} w_i[n], & \mathcal{H}_0, \\ h_i s[n] + w_i[n], & \mathcal{H}_1, \end{cases} \qquad E_i = \sum_{n=1}^{N} |x_i[n]|^2. \tag{3}$$

Here $i \in \{1, \ldots, K\}$ identifies an SU, n indexes samples within a slot, $s[n]$ is a PU signal, h_i is a sensing-link gain, and $w_i[n]$ denotes additive white Gaussian noise. The hypothesis $\mathcal{H}_h$ means $H_t = h$; the slot index is omitted from local quantities for readability. Equation (3) provides the physical motivation. The implemented simulator does not instantiate $s[n]$, h_i, or waveform samples. Instead, it directly samples the normalized energy distributions specified below. No modulation, carrier frequency, sampling rate, or fading distribution is inferred from this abstraction.

For a configured mean SNR in dB, denoted by $\bar{g}$, the simulator generates

$$G_{i,t} = \bar{g} + Z_{i,t}, \qquad Z_{i,t} \sim \mathcal{N}(0, \sigma_g^2), \qquad \gamma_{i,t} = 10^{G_{i,t}/10}, \quad \sigma_g = 1 \text{ dB}. \tag{4}$$

The disturbances are independent across SU indices and slots and independent of PU occupancy. Thus the linear SNR γ is positive and lognormal. The horizontal axis in the figures is $\bar{g} = \mathbb{E}[G]$, not $10 \log_{10} \mathbb{E}[\gamma]$. At a fixed $\bar{g}$, local sensing observations are conditionally independent across SUs given the PU state.

C. Gaussian Energy Model and Local Threshold

The default `paper` convention adopts the normalized Gaussian energy moments in [1, eq. (7)]:

$$\begin{aligned} E_i \mid \mathcal{H}_0, \gamma_i &\sim \mathcal{N}(N, 2N), \\ E_i \mid \mathcal{H}_1, \gamma_i &\sim \mathcal{N}\big(N(1+\gamma_i),\, 2N(1+2\gamma_i)\big). \end{aligned} \tag{5}$$

The second normal-distribution argument is the variance. Equation (5) is the exact sampling law of the implemented statistic-level simulator and an approximation to a waveform-level energy detector. The normalized energies are dimensionless; they are unrelated to battery energy in joules. The project uses $N \in \{200, 300, 400\}$, consistent with the large-sample regime adopted in [1]. These moment expressions should not be reinterpreted as a universal model for arbitrary complex I/Q signals.

The local threshold is an implementation choice that is not given explicitly in [1]. Using the equal-cost Bayesian density-comparison principle [7], let $f_h(e \mid \gamma)$ be the normal density in (5). The equal-density criterion uses equal prior and error weights locally. Setting $f_0(\lambda \mid \gamma) = f_1(\lambda \mid \gamma)$ and taking logarithms gives

$$\frac{(\lambda - N)^2}{4N} - \frac{\big(\lambda - N(1+\gamma)\big)^2}{4N(1+2\gamma)} = \frac{1}{2}\ln(1+2\gamma). \tag{6}$$

Solving the equal-density equation in (6) is an algebraic derivation for this project, not a threshold quoted from [1]. Multiplying by $4N(1+2\gamma)$, expanding, and dividing by 2γ yields $\lambda^2 - N\lambda - N^2\gamma/2 - N(1+2\gamma)\ln(1+2\gamma)/\gamma = 0$. The code uses its positive root,

$$\lambda_{i,t} = \frac{N}{2} + \sqrt{(1+2\gamma_{i,t})\left[\frac{N^2}{4} + N\frac{\ln(1+2\gamma_{i,t})}{\gamma_{i,t}}\right]}. \tag{7}$$

This is the upper equal-density crossing for the implemented one-threshold detector. A full likelihood-ratio test between unequal-variance normal densities on the entire real line also includes a lower, negative crossing. The code neither applies that two-tail rule nor clips Gaussian energy draws. The positive threshold preserves the conventional energy-detection form. Its evaluation uses `log1p` and the limit $\ln(1+2\gamma)/\gamma \to 2$ for small γ. At exactly zero SNR, the two distributions coincide and no threshold can distinguish them.

Following conventional threshold-based energy detection [3], [5], the reported local decision is

$$D_{i,t} = \mathbf{1}\{E_{i,t} \geq \lambda_{i,t}\} \in \{0, 1\}. \tag{8}$$

The threshold uses the realized local SNR in both hypotheses. Accordingly, this simulation assumes that such SNR information is available for threshold setting; no practical SNR estimator is implemented. The FC receives the resulting bits, not the SNR values or energies.

Evaluating the Gaussian upper tails of (5) at threshold (7) gives the following project-specific conditional probabilities [7]:

$$\begin{aligned} p_f(\gamma) &= Q\left(\frac{\lambda(\gamma) - N}{\sqrt{2N}}\right), \\ p_d(\gamma) &= Q\left(\frac{\lambda(\gamma) - N(1+\gamma)}{\sqrt{2N(1+2\gamma)}}\right), \end{aligned} \tag{9}$$

where $Q(a) = \Pr(U > a)$ for $U \sim \mathcal{N}(0,1)$. These are exact normal-tail expressions under (5). Population probabilities for the simulation additionally average (9) over (4). Substitution of $10^{\bar{g}/10}$ alone does not perform that averaging.

III. CNN-Based Cooperative Sensing

A. Dataset and Input Representation

Following the binary local-decision fusion representation of [1], one learning example corresponds to one slot. Its target is the simulated occupancy, and its input is a project-specific, lossless bipolar encoding of the K local bits:

$$\mathbf{X}^{(m)} = 2\,[D_1^{(m)}, \ldots, D_K^{(m)}] - \mathbf{1}, \qquad y^{(m)} = H^{(m)}. \tag{10}$$

The same transformation is applied to training, validation, and test data. A sample is reshaped to $1 \times K \times 1$ to match the convolutional layer interface. The input contains neither raw energy, SNR, previous occupancy, nor the target label. Encoding zero and one as minus one and plus one changes the numerical representation without adding information.

For each (K, N) configuration, separate generators produce 1,200 training and 300 validation slots at each of the 11 mean-SNR values from -16 to -6 dB. The resulting sets contain 13,200 and 3,300 examples, respectively. Independent seed assignments separate the generated sequences; observations within each sequence still inherit Markov label dependence. Identical binary feature patterns may naturally occur in different sets because the input alphabet has only 2^K elements. Such repetition alone is not evidence of data leakage.

B. Architecture and Decision Rule

The network retains the compact architecture of [1, Sec. 4.2]. Convolution and the ReLU nonlinearity follow the standard formulations in [8], [9]. It has K valid convolutional filters of size 1×2 and unit stride, followed by ReLU, a fully connected layer with two outputs, softmax, and classification. There is no pooling, recurrent layer, dropout, or batch normalization. For filter $r \in \{1, \ldots, K\}$ and position $j \in \{1, \ldots, K-1\}$,

$$z_{r,j} = w_{r,1} X_j + w_{r,2} X_{j+1} + b_r, \qquad a_{r,j} = \max(0, z_{r,j}). \tag{11}$$

Using the conventional softmax classifier [10], after flattening the activations into $\mathbf{a} \in \mathbb{R}^{K(K-1)}$, the two class logits and decision are

$$q_c = \mathbf{v}_c^{\mathsf{T}} \mathbf{a} + \beta_c, \qquad \widehat{p}_c = \frac{e^{q_c}}{e^{q_0} + e^{q_1}}, \qquad \widehat{H}_t = \arg\max_{c \in \{0,1\}} \widehat{p}_c. \tag{12}$$

Here c indexes the class, $\mathbf{v}_c$ and β_c are fully connected weights and biases, and $\widehat{p}_c$ is a model score conditional on the input. Probability calibration is not evaluated. The filters operate on neighboring SU indices; no physical spatial adjacency is assigned to those indices.

Table I gives the dimensions of the base network. Counting the actual convolutional weights and biases and the fully connected parameters in this implementation gives

$$3K + 2K(K-1) + 2 = 2K^2 + K + 2. \tag{13}$$

It equals 57, 212, and 467 for $K = 5, 10, 15$, respectively. Thus varying K also varies the network size.

TABLE I
Base CNN architecture for $K = 10$.

Layer	Output dimensions	Trainable parameters
Bipolar input	$1 \times 10 \times 1$	0
Valid convolution, ten 1×2 filters	$1 \times 9 \times 10$	30
ReLU and flattening	90	0
Fully connected	2 logits	182
Softmax and classification	2 scores and 1 label	0
Total		212

C. Learning and Model Selection

Let θ collect all trainable parameters and $\mathcal{W}$ denote the weights excluding biases. The mini-batch cross-entropy objective follows the standard softmax classification loss [10], while the added squared-weight penalty follows the MATLAB training configuration [11]. For a mini-batch $\mathcal{B}$, the data loss and regularized training objective are

$$\begin{aligned} \mathcal{L}_{\mathrm{CE}}(\theta; \mathcal{B}) &= -\frac{1}{|\mathcal{B}|} \sum_{m \in \mathcal{B}} \ln \widehat{p}_{y^{(m)}}(\mathbf{X}^{(m)}; \theta), \\ \mathcal{L}_{\mathrm{train}} &= \mathcal{L}_{\mathrm{CE}} + \frac{\lambda_{\mathrm{reg}}}{2} \sum_{W \in \mathcal{W}} \|W\|_F^2. \end{aligned} \tag{14}$$

where $\lambda_{\text{reg}} = 10^{-4}$ and $\|\cdot\|_F$ denotes the Frobenius norm. Adam [12] minimizes this objective with learning rate 0.001 and mini-batches of 128 examples. The convolutional weights use He initialization [13] and the fully connected weights use Glorot initialization [14]. Training data are shuffled every epoch. MATLAB's weight regularization and validation options are specified explicitly where exposed in the project [11].

Each configuration uses three random initializations on the same training and validation sets. A run has a maximum of 60 epochs; validation is evaluated every $\lfloor 13200/128 \rfloor = 103$ iterations. The patience of eight counts validation checks without improvement, not eight individual iterations. Each run returns its best validation-loss checkpoint. The code then recomputes unregularized validation cross-entropy and selects the lowest-loss candidate among the three runs. This procedure selects one network per configuration; it is not an ensemble. Test observations are not used by the selection function.

Five distinct networks are retained for $(K, N) \in \{(10, 300), (5, 300), (15, 300), (10, 200), (10, 400)\}$. The base network and its test outcomes are reused across all three studies, ensuring identical base curves rather than independently resimulated baselines.

IV. Simulation Results and Discussion

A. Setup and Evidence Base

The analyzed artifact set contains complete training and validation arrays for the five configurations, 15 training histories, five selected model records, per-repeat confusion counts, numerical summaries, and three result figures. Saved training metadata identify MATLAB 24.2.0.2712019 (R2024b); the configured execution environment is CPU and training uses Deep Learning Toolbox. Table II summarizes the experiment settings. Numerical findings below describe these archived simulations. For this manuscript, the supplied outputs were postprocessed and checked for consistency; MATLAB training and inference were not independently rerun.

TABLE II
Simulation and Learning Settings.

Item	Value
PU transitions and stationary occupancy	$p_{01} = p_{10} = 0.2$; $\pi_0 = \pi_1 = 0.5$
Mean SNR grid; SNR standard deviation	$-16 : 1 : -6$ dB; 1 dB
Base configuration	$K = 10$, $N = 300$
Cooperation study	$K \in \{5, 10, 15\}$, $N = 300$
Sample-count study	$N \in \{200, 300, 400\}$, $K = 10$
Energy convention	Gaussian statistics in (5), `paper` mode
Training / validation per configuration	13,200 / 3,300 slots
Test budget	5,000 slots per SNR per repeat; 4 repeats
Optimizer; mini-batch; L2 coefficient	Adam; 128; 10^{-4}
Learning rate; maximum epochs	0.001; 60
Validation frequency; patience	103 iterations; 8 validation checks
Initializations; selection	3; minimum validation cross-entropy
Seed bases: data / test / initialization	202610 / 320261105 / 920261105

In the method comparison, the CNN, strict-majority baseline, and single-user detector receive the same test observations and labels in each repeat. For the other configurations, independently seeded test sequences are generated. Changing N includes retraining; it does not test a fixed model trained at $N = 300$ on unseen sample-count settings. Likewise, every tested mean SNR is represented in the training grid. The experiments assess held-out observations from the specified simulator, not extrapolation to unseen SNRs or real radio measurements.

B. Baselines and Performance Measures

Hard-decision voting is a standard cooperative-fusion comparator [4], [15]. Let $V_t = \sum_{i=1}^{K} D_{i,t}$ be the vote count. The project-specific strict-majority and single-user baselines are

$$\widehat{H}_{\text{SM},t} = \mathbf{1}\{V_t \geq \lfloor K/2 \rfloor + 1\}, \qquad \widehat{H}_{\text{SU1},t} = D_{1,t}. \tag{15}$$

Strict majority is the rule labeled `HALF` in the code. For $K = 10$, it requires six active votes. The half-voting definition in [1] and the diagnostic at-least-half rule instead use $V_t \geq \lceil K/2 \rceil$, requiring five votes. The optimum rule developed by Chair and Varshney [16], which is also discussed in [1], is outside the selected project comparisons, so no conclusion of near-optimal fusion performance is drawn.

For one test sequence, let TP, FN, FP, and TN be the true-positive, false-negative, false-positive, and true-negative counts. The estimated conditional probabilities and the reference sensing-error sum are

$$\widehat{P}_d = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \qquad \widehat{P}_f = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}}, \qquad \widehat{P}_e = \widehat{P}_f + 1 - \widehat{P}_d. \tag{16}$$

These detection, false-alarm, and sensing-error definitions follow [1, eqs. (1), (2), and (28)]. Since $P_e \in [0, 2]$, it must not be identified with the ordinary classification-error probability. If T is the number of tested slots and $\widehat{\pi}_h$ is the observed fraction in class h, the empirical classification error follows directly from the confusion counts as

$$\widehat{P}_{\mathrm{err}} = \frac{\mathrm{FP}+\mathrm{FN}}{T} = \widehat{\pi}_0 \widehat{P}_f + \widehat{\pi}_1 (1 - \widehat{P}_d). \tag{17}$$

At balanced population priors, $P_{\mathrm{err}} = P_e/2$; finite Markov sequences need not satisfy this equality exactly. For each SNR, a reported metric is the mean of four per-repeat estimates. If m_r is an estimate from repeat r, its mean and sample standard deviation are

$$\overline{m} = \frac{1}{4}\sum_{r=1}^{4} m_r, \qquad s_m = \sqrt{\frac{1}{3}\sum_{r=1}^{4}(m_r - \overline{m})^2}. \tag{18}$$

Error bars in Figs. 2 and 3 show $\overline{P}_e \pm s_{P_e}$. They describe test-set variability for a fixed selected network; they are neither 95% confidence intervals nor uncertainty across retraining runs. Recalculation of all 396 method-condition-repeat records confirms the saved metric formulas and their summary means and standard deviations to numerical precision. The three base curves match exactly.

C. Method Comparison and the Effect of Ties

Figure 1 reports detection and false-alarm probabilities for $K = 10$ and $N = 300$. Across the tested SNR grid, the CNN has higher detection probability than strict majority, but also a higher false-alarm probability. Its lower error sum is therefore a trade-off between reduced misses and additional false alarms, rather than improvement in both conditional metrics.

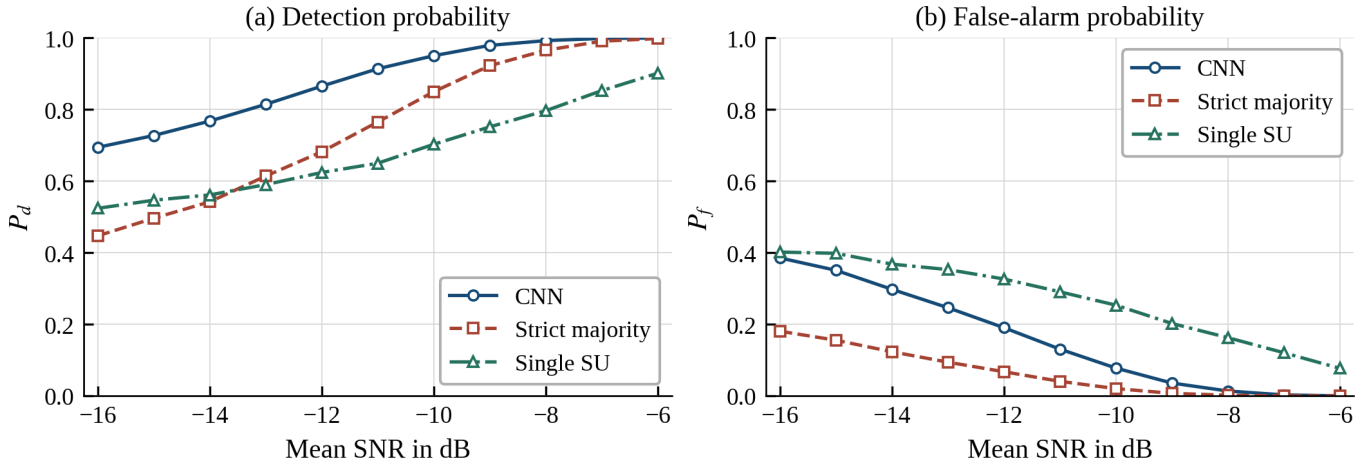


Fig. 2. Detection and false-alarm probabilities at $K = 10$ and $N = 300$. Points are means over four test repeats. Strict majority denotes the code's `HALF` rule and requires at least six active votes.

At $\bar{g} = -12$ dB, $(\overline{P}_d, \overline{P}_f, \overline{P}_e)$ is $(0.8653, 0.1907, 0.3254)$ for the CNN, $(0.6820, 0.0677, 0.3857)$ for strict majority, and $(0.6241, 0.3267, 0.7025)$ for SU1. Cooperation substantially reduces the error sum relative to the single local decision under this configuration. The CNN's false-alarm probability of approximately 0.19 means that about 19% of actually idle slots are rejected at this operating point.

The additional supplied audit is decisive for interpreting the CNN-versus-voting difference. All 1,024 possible binary inputs of the base network were grouped by vote count. Every pattern with at most four active votes is classified inactive; every pattern with at least six active votes is classified active. Of the 252 patterns with exactly five votes, 250 are classified active. The CNN therefore differs from at-least-half voting on only two possible inputs. On the 220,000 observations of the base test grid, the two rules disagree on 181 decisions, or approximately 0.0823%.

Table III reports the arithmetic mean of $\overline{P}_e$ over the 11 configured SNR points. This grid average weights the SNR points equally and is not a deployment-weighted performance estimate. The almost identical CNN and at-least-half values show that the apparent CNN gain over strict majority is closely associated with tie handling. The small observed difference between

TABLE III
BASE-MODEL SENSING-ERROR SUM AVERAGED UNIFORMLY OVER THE TESTED SNR GRID.

Decision method	Mean of $\overline{P}_e$
CNN	0.275433
Strict majority, at least 6 of 10	0.310837
At least half, at least 5 of 10	0.275170
Single-user energy detection	0.586713

CNN and at-least-half voting has not been tested for statistical significance and gives no evidence of a useful CNN advantage over that rule.

This behavior is consistent with the simulator's statistical symmetry. At a fixed $\bar{g}$, all SU decisions have the same conditional distribution and are independent given occupancy. Consequently, the probability of a binary vector under either hypothesis depends on the vector through its vote count (a direct factorization under the model; cf. classical distributed detection [15], [16]). Pooling the SNR settings can introduce conditional dependence through their common SNR setting, but preserves this dependence of the vector likelihood on vote count alone. A count-based rule is therefore a strong comparator. The convolution can approximate such a rule, but these experiments do not establish that convolutional feature extraction is necessary.

At -6 dB, the CNN's mean error sum is approximately 4.97×10^{-4}, while its empirical classification error is 0.00025. Across the four base repeats, the latter corresponds to two misses and three false alarms among 20,000 slots. These rare-error observations are informative but do not establish zero population error.

D. Effect of the Number of Cooperating Users

Figure 2 holds $N = 300$ and varies K. The reported mean error sums are ordered as $K = 15 < K = 10 < K = 5$ throughout the evaluated grid. At -12 dB, the values are 0.2455, 0.3254, and 0.4806 for $K = 15, 10, 5$, respectively. At -10 dB, they are 0.0700, 0.1275, and 0.2697.

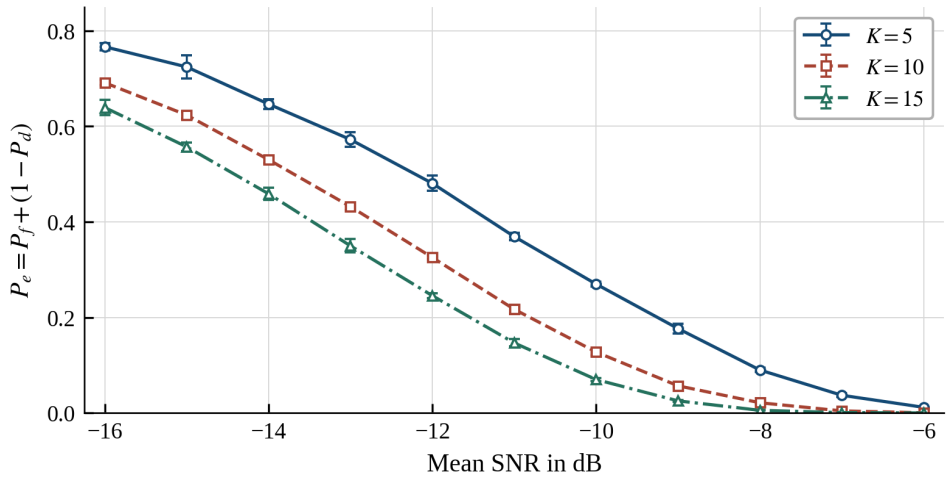


Fig. 3. CNN sensing-error sum for $K \in \{5, 10, 15\}$ with $N = 300$. Error bars are one sample standard deviation across four test repeats for each fixed selected network.

Under conditional independence, additional users provide more binary observations of the same occupancy state. If their local decisions are informative, the aggregated vote distribution can separate the hypotheses more strongly. This explains why increased cooperation can improve classification in the adopted model. Nevertheless, this experiment changes both sensing diversity and model capacity: the CNN parameter count increases from 57 to 467. It does not isolate a pure user-count effect at constant network size, nor does it prove the same gain under correlated sensing channels.

The shared $K = 10, N = 300$ result in this figure is exactly the CNN result in Fig. 1. The error bars measure new test-sequence variation, while initialization variability is represented only by the saved training records and is not included in those bars.

E. Effect of the Nominal Sensing Sample Count

Figure 3 fixes $K = 10$ and varies N. At -12 dB, the mean error sum falls from 0.4155 for $N = 200$ to 0.3254 for $N = 300$ and 0.2758 for $N = 400$. At -10 dB, the corresponding values are 0.2073, 0.1275, and 0.0957. The same ordering is observed across the tested grid.

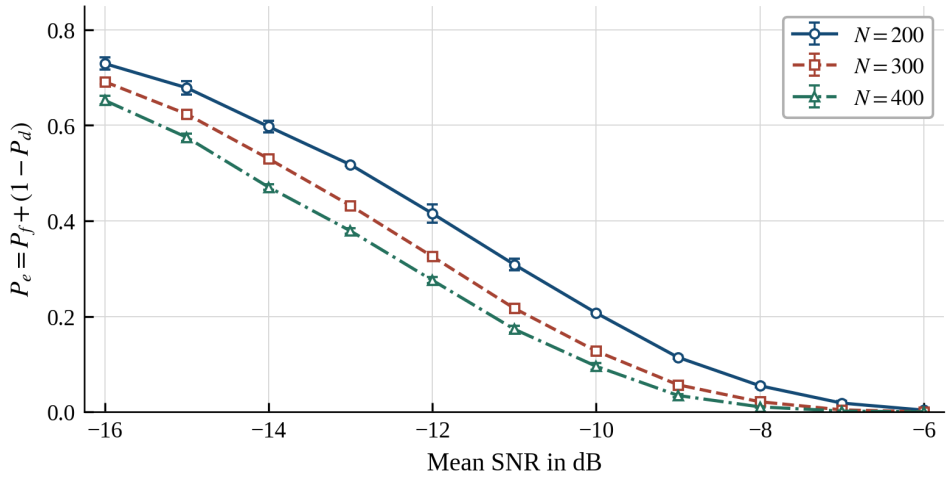


Fig. 4. CNN sensing-error sum for $N \in \{200, 300, 400\}$ with $K = 10$. A separate network is trained for each N; error bars show one test sample standard deviation.

Equation (5) explains this trend at the statistic level. For fixed γ, the conditional mean difference is $N\gamma$, whereas the noise-only standard deviation is $\sqrt{2N}$. Their ratio is $\gamma\sqrt{N/2}$, which increases with N. Thus the modeled energy distributions become more distinguishable, and the binary decisions supplied to the FC become more reliable. This interpretation concerns the Gaussian generator; it is not evidence that the program processes actual blocks of N received I/Q samples.

At the high-SNR end, all three error sums become small and finite test counts limit the precision of comparisons. For example, the $N = 400$ setting at -6 dB contains one miss and two false alarms in its 20,000 evaluated slots. No sampling rate or sensing-time budget is specified, so increasing N cannot be translated into a measured delay, energy cost, or throughput trade-off in this project.

F. Limitations and Scope of the Findings

The results depend on synthetic Gaussian energy statistics, a specified SNR distribution, conditionally independent and identically distributed SUs at each fixed nominal SNR, ideal reporting, and SNR-aware local thresholds. Channel correlation, reporting errors, noise-power uncertainty, receiver mismatch, and measured radio data are not evaluated. The Gaussian approximation is adopted from [1] rather than validated against a waveform simulator here. In practical sensing, noise-power uncertainty can impose fundamental performance limitations [17]; such uncertainty was not simulated in this project.

The trained CNN consumes current-slot binary decisions and discards raw energy information. It does not use Markov history, and its architecture is not permutation invariant by construction. The base-model audit shows small order-dependent decision differences among tied vote patterns. These facts limit interpretations based on temporal learning or physical spatial feature extraction.

The training and test sets use separate seeds, and code-level model selection uses validation alone. This supports within-simulator held-out evaluation. It does not establish robustness to distribution shift, account for all variation across retraining, or independently certify the complete history of manual experimental choices. The saved MATLAB model objects were not re-executed during manuscript preparation. The recorded results, confusion counts, and metadata are consistent, but an independent end-to-end reproduction remains a separate validation step.

V. Conclusion

This project reimplemented the CNN-based cooperative spectrum-sensing component of [1] using synthetic binary local decisions and an explicit validation-based training protocol. The archived experiments show lower sensing-error sums for the cooperative CNN than for the single-user detector and the implemented strict-majority baseline across the evaluated SNR grid. Increasing the number of cooperating users and the nominal sensing sample count also reduces the observed error sum within the tested configurations.

The diagnostic audit substantially qualifies the CNN-versus-voting conclusion. The base CNN behaves almost like at-least-half voting, and that simple rule has essentially the same grid-averaged performance. The project therefore demonstrates a functioning learned fusion pipeline and the benefit of cooperative information under its assumptions, while not establishing an intrinsic advantage of CNNs over well-specified voting. Future work could evaluate measured or waveform-generated data, imperfect SNR knowledge, sensor dependence, and other information representations, with any new claims supported by separate experiments.

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